

NYC

Energy & CO₂

Analytics



DATA-DRIVEN INSIGHTS. SMARTER DECISIONS.
GREENER FUTURE.

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THE PROBLEM



Why We Started This Project



High Energy Consumption

NYC buildings consume vast amounts of energy every year, leading to high operational costs and inefficient resource use.



Rising CO₂ Emissions

Building emissions are a major contributor to climate change, impacting air quality and public health across the city.



LL97 Law & Penalty Risks

Local Law 97 sets strict emission limits. Many buildings are at risk of heavy penalties if they fail to meet the required targets.



Complex Data, Limited Clarity

Energy and emissions data is scattered and complex, making it hard to identify priorities and choose the best solutions.



We saw the challenge.

Our mission is to turn complex data into clear insights and actionable solutions for a greener, smarter NYC.



OUR SOLUTION

Turning complex data into clear actions

To tackle NYC's energy, emissions, and penalty challenges, we built an end-to-end data solution that transforms raw data into **insights** and supports **smarter decisions**.



01

CLEAN & PREPARE

We collected raw NYC building data and performed extensive cleaning to remove errors, duplicates, and inconsistencies, ensuring reliable and high-quality data.



02

STRUCTURE WITH SQL

We designed a relational database and structured the data in **SQL Server** to enable efficient storage, strong relationships, and powerful querying.



03

ANALYZE & VISUALIZE

We built interactive dashboards in **Excel**, **Tableau**, and **Power BI** to explore trends, compare scenarios, and uncover key insights.



04

BUILD SMART PLATFORM

We developed a Python-powered platform with dashboards, scenario-based solutions, an **AI chatbot**, and an **AI model** to help users explore data and choose the best actions.



05

DRIVE BETTER DECISIONS

Our solution delivers clear insights and recommendations to help stakeholders reduce emissions, minimize penalties, optimize investments, and move toward a **greener NYC**.



DATASET OVERVIEW

A comprehensive dataset containing **energy consumption** and **emissions** information for buildings across New York City.



SOURCE

NYC Open Data

NYC Building Energy and
Emissions Performance
(Local Law 84 & 97)

RECORDS

29,842

Raw Records

11,640

Final Cleaned Records

FEATURES

28+

Key Features

Including energy, emissions,
property, building, and
operational attributes.

GRANULARITY

Property Level

Each record represents
one property (building)
in NYC.

DATA DESCRIPTION

Properties are linked to location, building type, gross floor area, construction status, and metering information.

Energy metrics describe consumption and performance; emission metrics translate energy behavior into carbon impact.

Penalty and alert fields support risk analysis and prioritization for action.



DATA CLEANING PIPELINE

From raw LL97 property records to a clean, analysis-ready dataset

- 01 | Standardizing Null Values
- 02 | Filtering Test Properties
- 03 | LL97 Size Threshold Filter
- 04 | New York City Boundaries Filter
- 05 | Data Integrity & Alerts Filter
- 06 | Penalty Outliers Filter
- 07 | Zero Energy Use Filter
- 08 | Remove Duplicate Property IDs

CLEANING SUMMARY

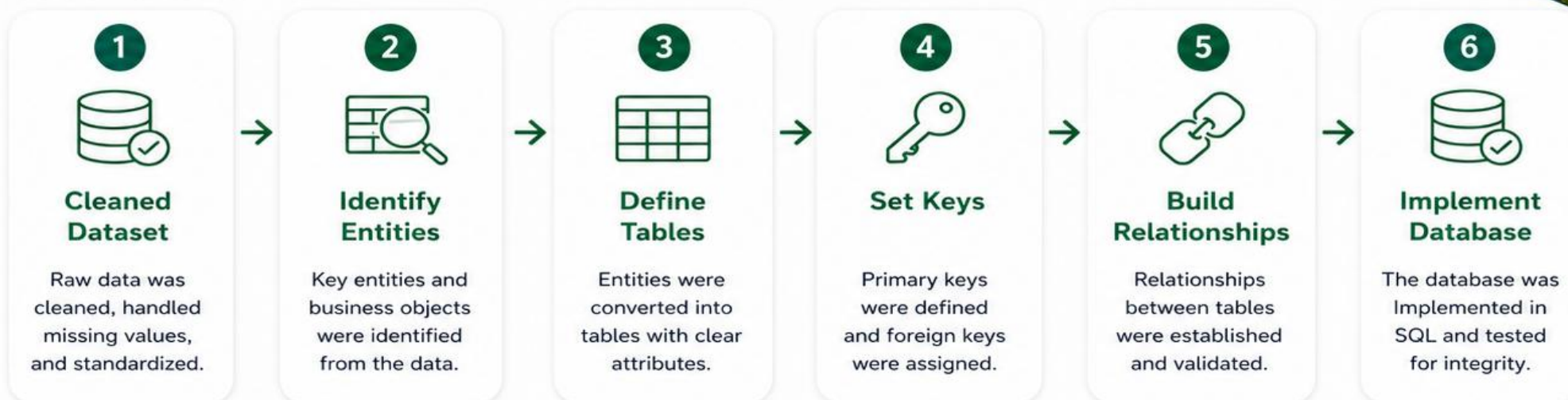
Raw Dataset	29,842
Final Dataset	11,640
Removed Records	18,202
Overall Reduction	61.0%

From Clean Data to Structured Database

Before building our SQL database, we cleaned and prepared the raw dataset to ensure accuracy, consistency, and reliability.

These steps transform raw data into a well-structured, relationship-driven database that supports powerful analysis and decision making.

OUR APPROACH



Our Goal

To build a reliable, scalable, and well-structured SQL database that supports accurate analysis, efficient querying, and data-driven decision making.



Reliable Data



Better Insights

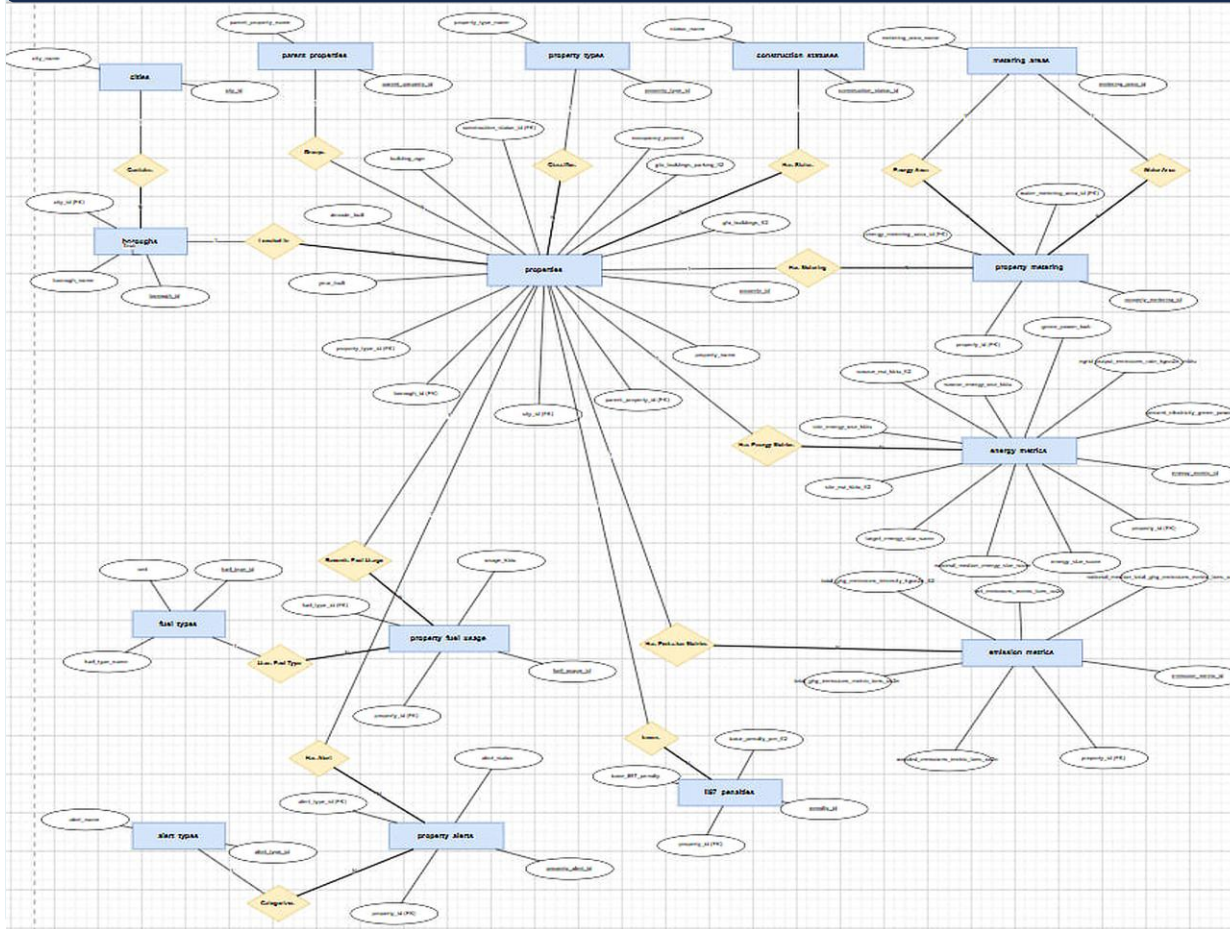


Smarter Decisions



SQL Database: ERD

ERD Overview



Notes

This ERD represents the database blueprint of the project. It shows the main entities, their attributes, and the relationships between tables. The diagram helps organize project data, reduce duplication, and support accurate SQL queries for dashboards, AI models, and scenario analysis.



SQL Schema & Tables

1	properties													
2	property	property	year	decad	build	occup	gfa_b	gfa_b	paren	city_id	boroug	proper	constr	
3														
4														
5	parent_properties													
6	parent	parent_pr												
7														
8														
9	cities													
10	city_id	city_name												
11														
12														
13	boroughs													
14	borough	borough_n	city											
15														
16														
17	property_types													
18	property	property												
19														
20														
21	construction_statuses													
22	construc	status_n												
23														
24														
25	metering_areas													
26	metering	metering												
27														
28														
29	property_metering													
30	property	property	energ	water										
31														
32														
33	energy_metrics													
34	energy	energy_st	natio	targe	site	site	source	source	green	egrid	percer	Column		
35														
36														
37	emission_metrics													
38	emission	avoided_e	total	total	net_e	natio	prope							
39														
40														
41	1197_penalties													
42	penalty	base_1197	base	prope										
43														
44														
45	fuel_types													
46	fuel_ty	fuel_type	unit											
47														
48														
49	property_fuel_usage													
50	fuel_us	usage_kbt	prope	fuel										
51														
52														
53														
54														
55	alert_types													
56	alert_t	alert_n												
57														
58	property_alerts													
59	property	property	alert	alert										
60														



Schema Overview

The schema organizes the cleaned dataset into related tables, where each table stores a specific type of information such as properties, cities, boroughs, energy metrics, emissions, fuel usage, and alerts.



Clear structure

Each table has a specific purpose.



Reduced duplication

Repeated values are stored in separate lookup tables.



Ready for SQL

The schema was used to build the final relational database.

Important SQL Queries

```
File Edit View Query Git Project Tools Extensions Window Help Search
New Query
master
Execute
carbon_heist...\beroo (58) X

1  -- Create all for Database
2  -- Create all for Database
3  -- Create all for Database
4  -- Create all for Database
5  IF NOT EXISTS (SELECT * FROM sys.databases WHERE name = 'TN_CARBON_HEIST')
6  BEGIN
7  CREATE DATABASE TN_CARBON_HEIST;
8  END
9  GO
10
11 USE TN_CARBON_HEIST;
12 GO
13
14 -- 1. Lookup (Dimension)
15 -- 1. Lookup (Dimension)
16 -- 1. Lookup (Dimension)
17
18 CREATE TABLE [CITY] (
19     [city_id] INT PRIMARY KEY,
20     [city_name] NVARCHAR(100) NOT NULL UNIQUE
21 );
22 GO
23
24 CREATE TABLE [BOROUGH] (
25     [borough_id] INT PRIMARY KEY,
26     [borough_name] NVARCHAR(100) NOT NULL UNIQUE,
27     [city_id] INT NOT NULL,
28     FOREIGN KEY ([city_id]) REFERENCES [CITY]([city_id]) ON DELETE NO ACTION
29 );
30 GO
31
32 CREATE TABLE [PROPERTY_TYPES] (
33     [property_type_id] INT PRIMARY KEY,
34     [property_type_name] NVARCHAR(100) NOT NULL UNIQUE
35 );
36 GO
37
38 CREATE TABLE [CONSTRUCTION_STATUSES] (
39     [construction_status_id] INT PRIMARY KEY,
40     [status_name] NVARCHAR(100) NOT NULL UNIQUE
41 );
42 GO
43
44 CREATE TABLE [FUEL_TYPES] (
45     [fuel_type_id] INT PRIMARY KEY,
46     [fuel_type_name] NVARCHAR(100) NOT NULL UNIQUE
47 );
48 GO
49
50 CREATE TABLE [FUEL_UNITS] (
51     [fuel_unit_id] INT PRIMARY KEY,
52     [unit_name] NVARCHAR(50) NOT NULL UNIQUE,
53     [unit_conversion] DECIMAL(18,4)
54 );
55 GO
56
57 CREATE TABLE [LAST_DATES] (
58     [last_date_id] INT PRIMARY KEY,
59     [last_update_date] DATE NOT NULL
60 );
61 GO
62
```

55 % No issues found

Connected. (1/1) (local) (17.0 RTM) BEK\Dber

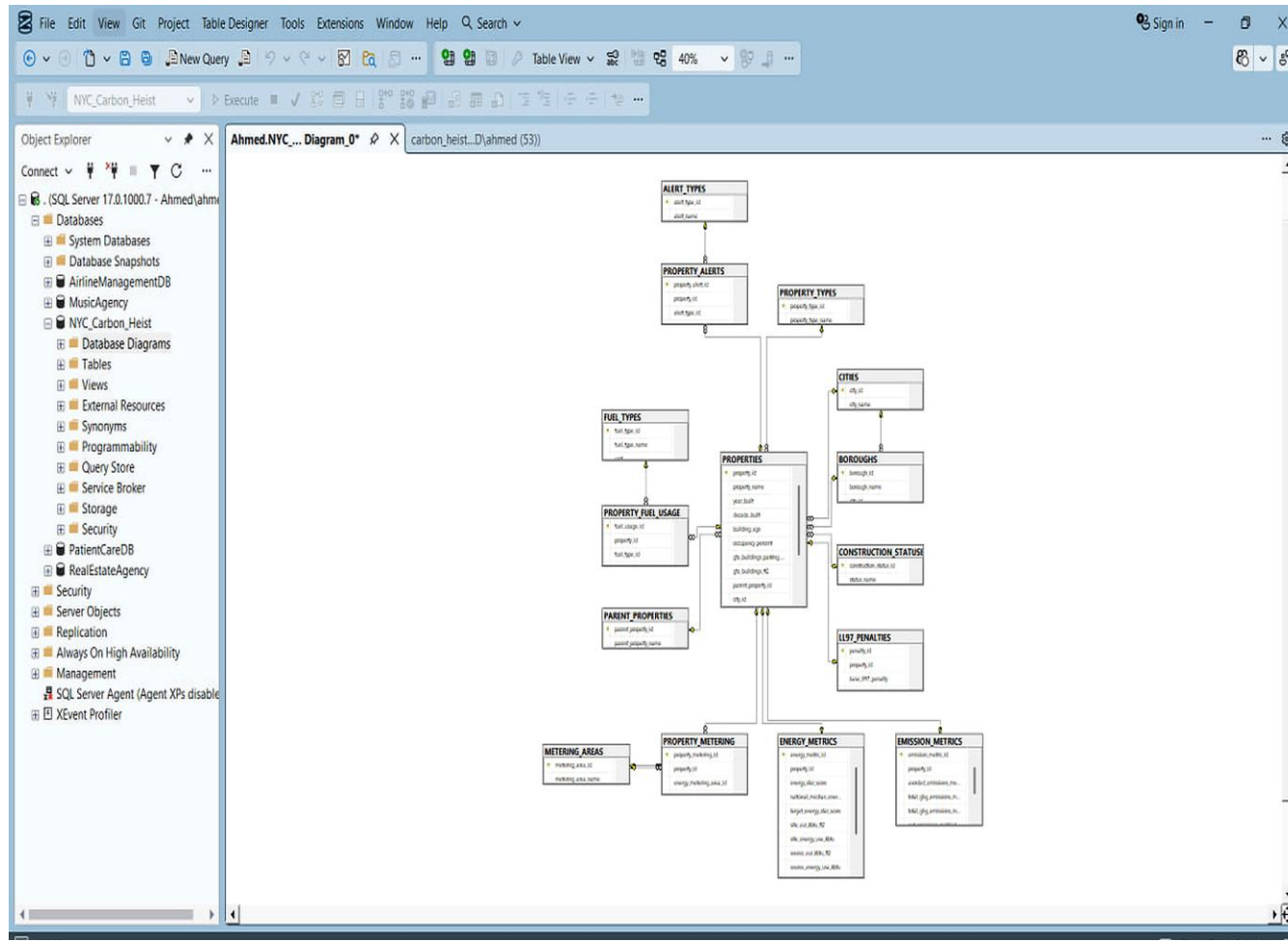


Note

These SQL queries were used to extract, filter, and summarize the key data needed for analysis and dashboard development.



Connected SQL Database



Note

This database was implemented to store the cleaned data in connected tables, ensuring consistency, reducing duplication, and enabling efficient querying for analysis and dashboards.





From SQL Data to Excel Dashboards

Now we turn our structured data into **interactive insights** using Excel dashboards.



SQL Database



Excel Dashboards



Key Metrics

Track essential KPIs such as penalties, savings, CAPEX, payback, and portfolio performance.



Interactive Analysis

Filters and comparisons enable deep exploration across properties, scenarios, and factors.



Scenario Evaluation

Compare intervention strategies and assess their impact on cost, savings, and emissions.



Decision Support

Identify priorities and select the most effective actions based on data-driven insights.



The Excel dashboards transform cleaned and structured data into clear business **insights** and **actionable decisions**.



EXCEL Dashboards



DASHBOARD PURPOSE

This dashboard summarizes the full decarbonization strategy across the portfolio. It compares total penalty risk, projected savings, and payback periods to support investment decisions.



DASHBOARD PURPOSE

This dashboard identifies the key factors driving emissions and penalty exposure. It compares building age, borough, area, property type, and Energy Star performance.



DASHBOARD PURPOSE

This dashboard presents a targeted retrofit strategy focused on quick, high-impact energy improvements. It compares penalty risk, project CAPEX, and projected savings before and after implementation.



DASHBOARD PURPOSE

This dashboard evaluates system optimization without major equipment replacement. It shows how improving existing operations can reduce emissions, penalties, and operating inefficiencies.



WET Systems Scenario

Penalty Risk Oer Year

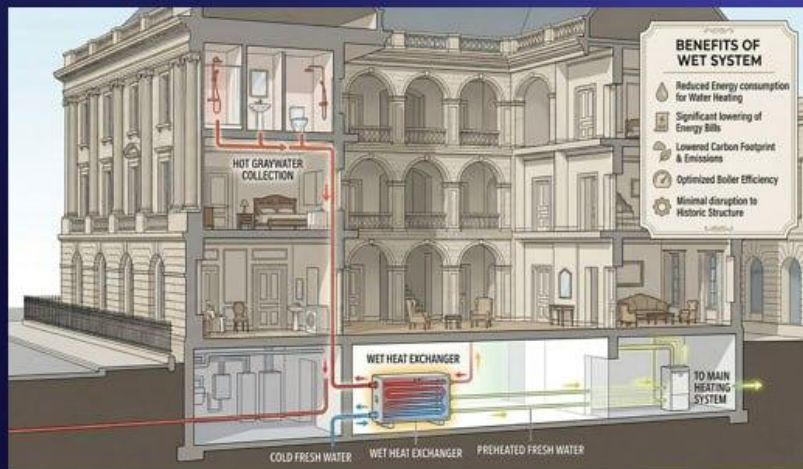
\$306.0M

Total Project CAPEX

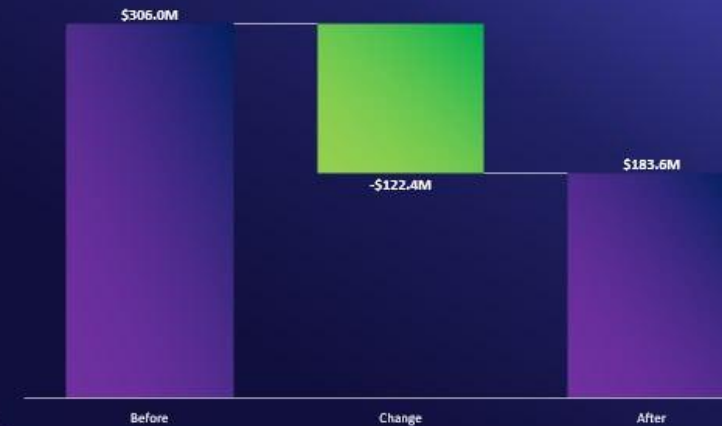
1.50B

Fine Savings Per Year

\$122.4M



WET Systems Penalty Reduction Bridge



DASHBOARD PURPOSE

This dashboard analyzes upgrades to wet/heating systems, especially in older buildings. It highlights how system improvements can reduce energy waste, emissions, and future penalties.



1960s Smart Scale Scenario

Penalty Risk Per Year

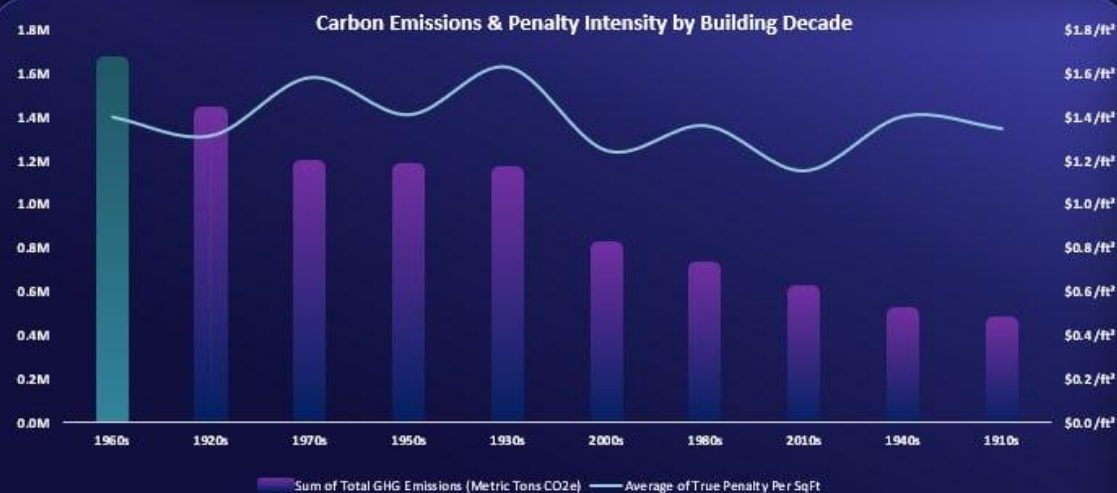
\$306.0M

Total Project CAPEX

\$785.2M

Fine Savings Per Year

\$88.0M

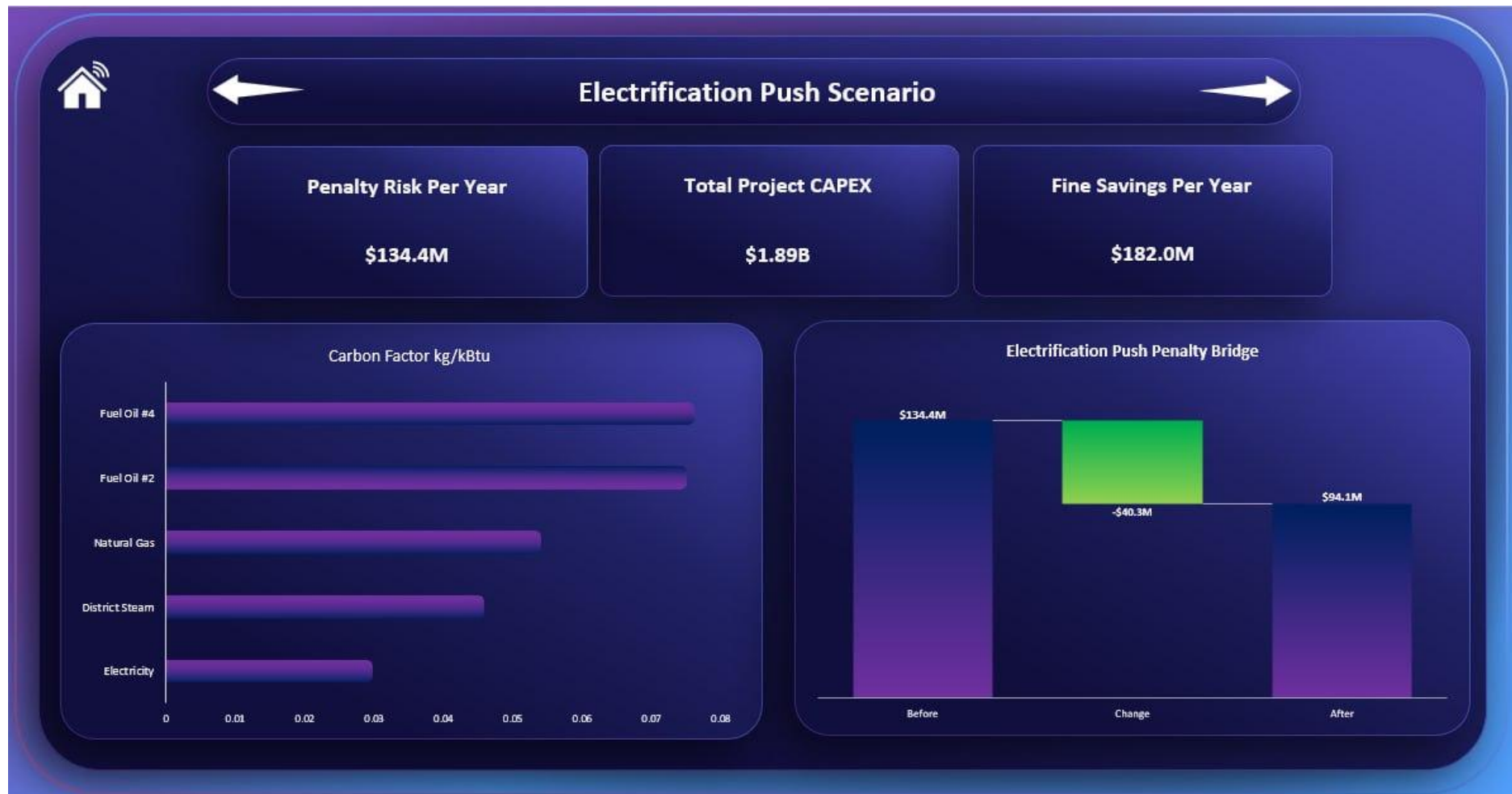


1960s Smart Scale Penalty Bridge



DASHBOARD PURPOSE

This dashboard focuses on 1960s buildings as a priority intervention group. It measures how smart-scale improvements can reduce penalties and improve financial outcomes.



DASHBOARD PURPOSE

This dashboard evaluates the impact of shifting building systems from fossil fuels to electricity. It shows how electrification affects carbon emissions, penalty risk, investment cost, and savings.

NEXT SECTION →

Tableau Dashboards

Turning data into powerful visual insights

Our Tableau dashboards provide an **interactive, data-rich** view of NYC's energy efficiency and emissions **performance** across boroughs, properties, and energy sources.



Interactive Visuals

Dynamic charts and maps to explore key energy and emissions metrics.



Geographic Insights

Borough-level analysis to identify high-impact opportunities.



Multi-Dimensional Analysis

Compare emissions, intensity, avoided emissions, and more across multiple dimensions.



Data-Driven Decisions

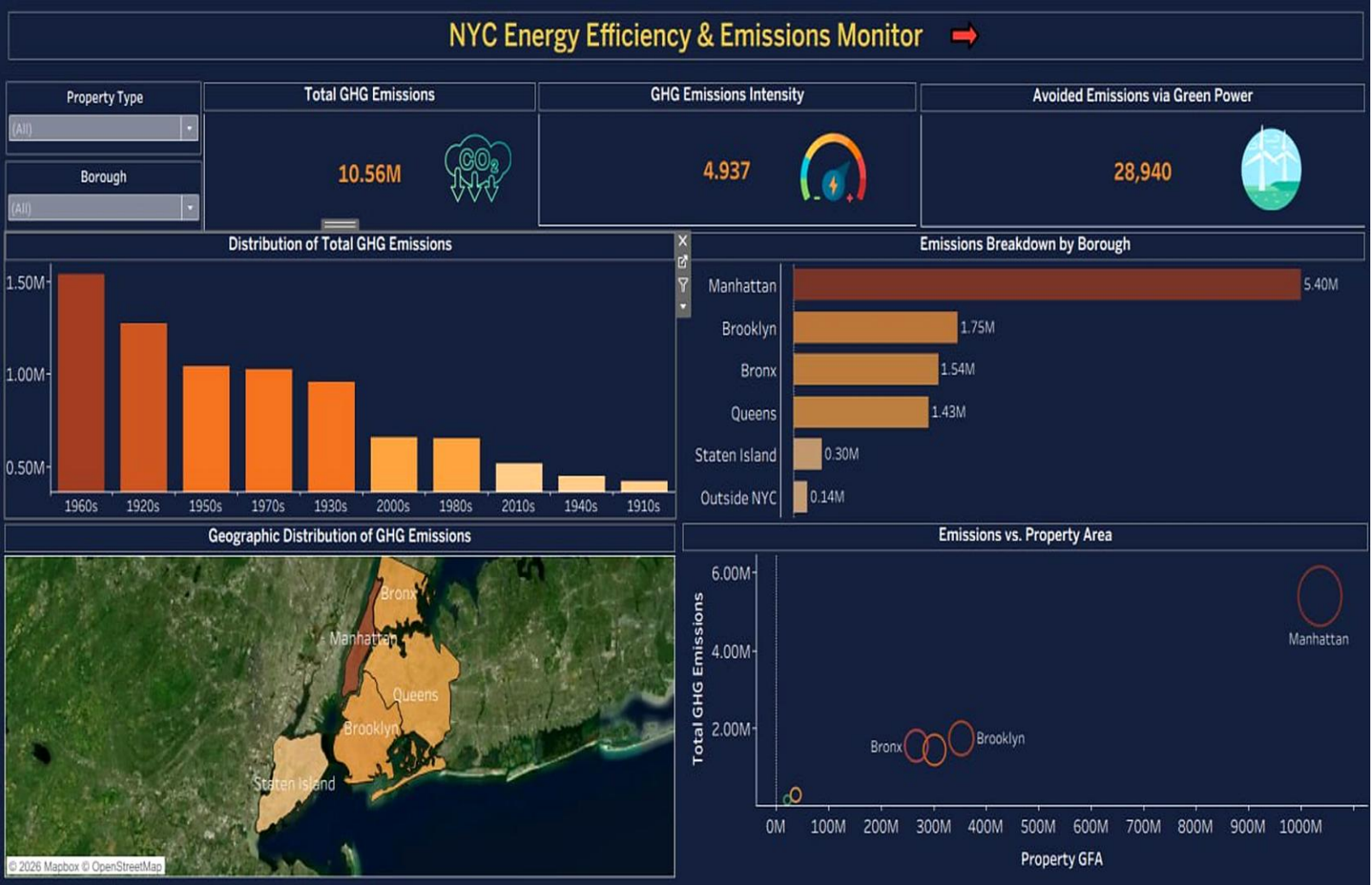
Visual insights that support smarter strategies and measurable impact.



Exploring deeper.
Analyzing smarter.
With **Tableau**.



Tableau Dashboards



Dashboard Overview

High-level summary of key energy and emissions metrics.



Total GrtG Emissions

Displays the overall greenhouse gas emissions across all properties.



GHG Emissions Intensity

Shows the average emissions intensity (kgCO₂e/ft²) across the portfolio.



Avoided Emissions via Green Power

Represents emissions avoided through the use of renewable energy.



Distribution of Total GHG Emissions

Illustrates total emissions distribution across building construction decades.



Emissions Breakdown by Borough

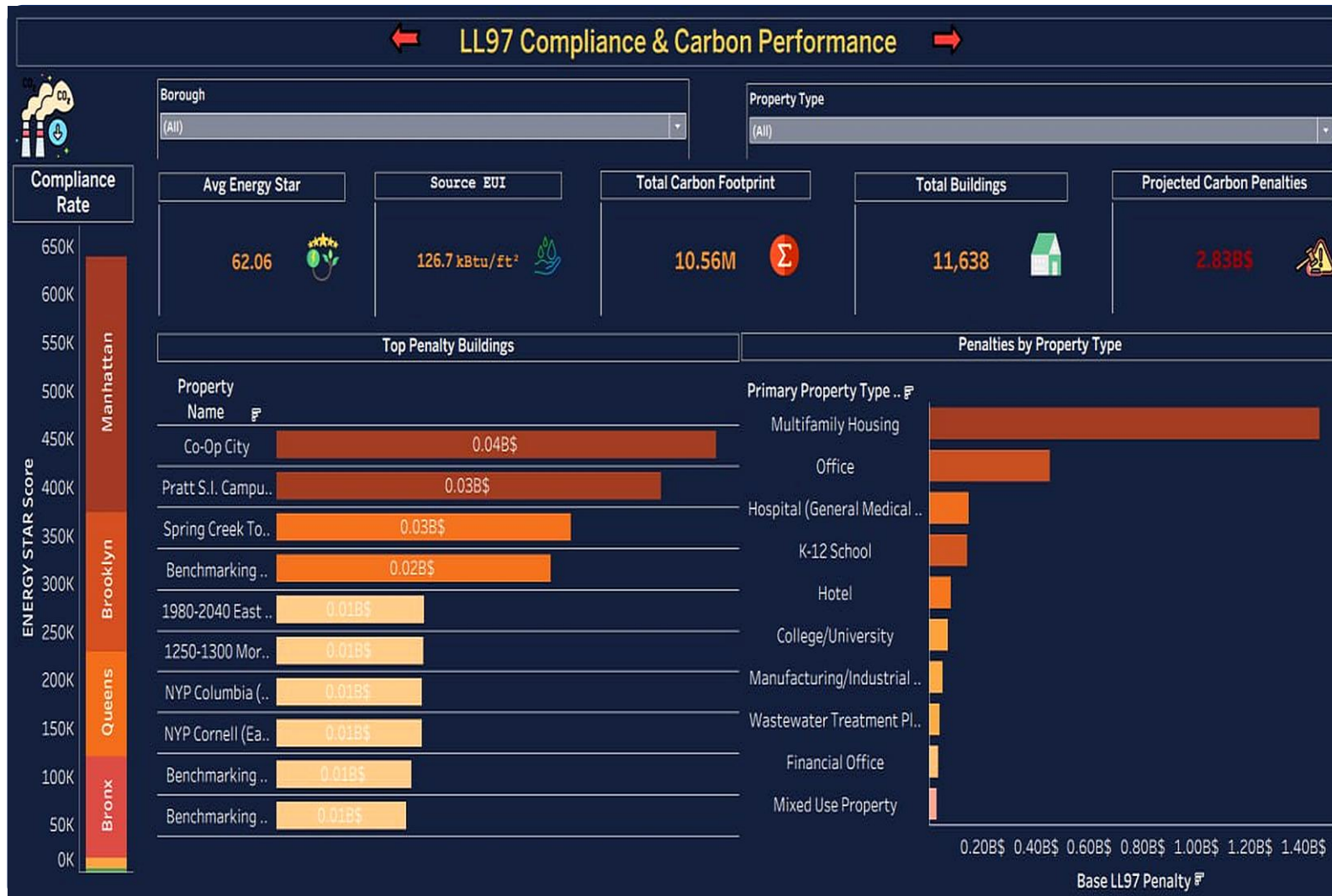
Compares total emissions contributed by each borough.



Geographic Distribution

Heat map showing spatial distribution of emissions across NYC boroughs.





Dashboard Overview



This dashboard provides a quick view of LL97 compliance, energy efficiency, carbon emissions, and projected penalty risk across NYC buildings.

It helps identify high-risk buildings and property types that require priority action.



Decarbonization Roadmap

Projected Carbon Penalties	Total Buildings	Penalized Buildings	Avg Penalty/sqft	Total Emissions
2.83B\$ 	11,638 	7,309 	1.30\$ 	10.56M 

LL97 Mitigation Roadmap



Scan to Explore the Live Interactive C-Suite Portal





CARBON COST & SAVINGS ANALYSIS

POWER BI DASHBOARDS

Transforming data into clear, interactive insights for smarter energy and carbon management.



Visualize
Key Metrics



Track Trends
& Performance



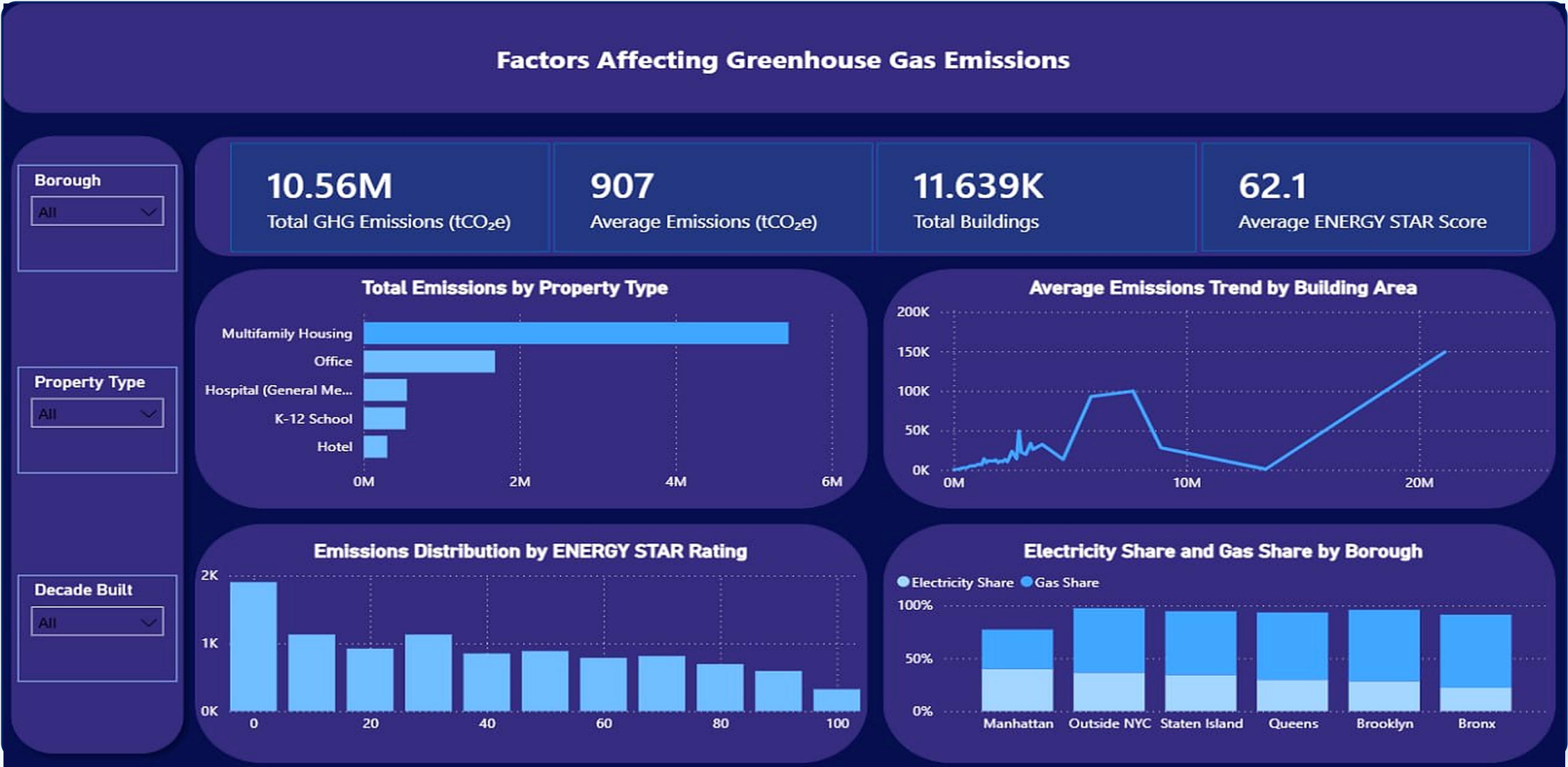
Monitor Savings
& Penalties



Drive Better
Decisions



Power BI Dashboards



Factors Affecting Greenhouse Gas Emissions

Borough

All

Property Type

All

Decade Built

All

10.56M

Total GHG Emissions (tCO₂e)

Manhattan

Highest Emitting Borough

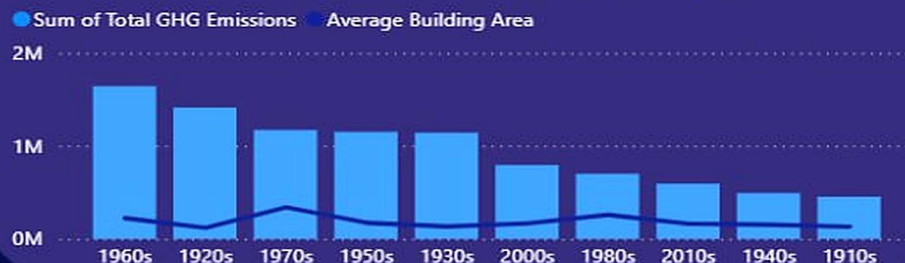
11.639K

Total Buildings

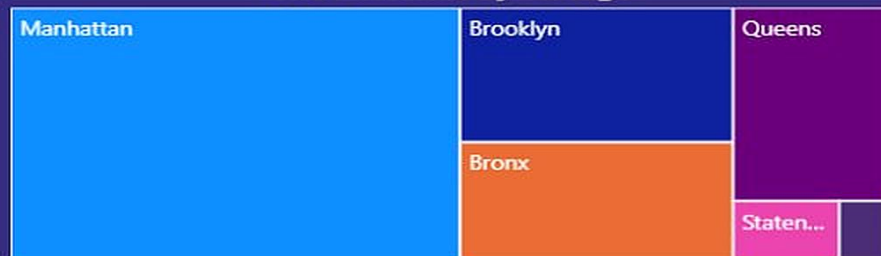
62.1

Average ENERGY STAR Score

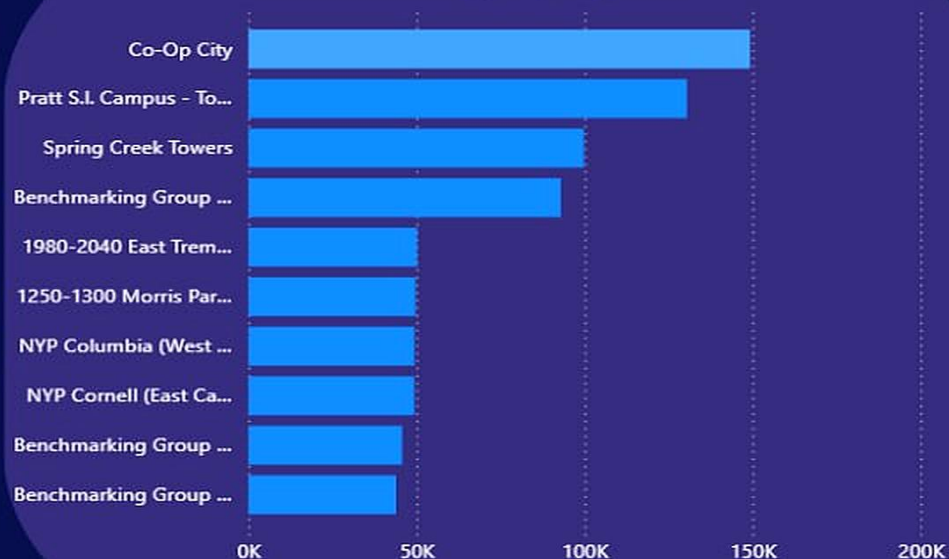
GHG Emissions vs Building Area by Construction Decade



GHG Emissions by Borough



Total Emissions by Property Name



Carbon Cost & Savings Analysis

10.56M

Total GHG Emissions (tCO₂...)

2.83bn

Estimated Carbon Penalty

641M

Total Fine Savings

0.23

Highest Penalty Proper...

62.1

Average ENERGY STAR Score

Borough

All

Property Type

All

Decade Built

All

Estimated Carbon Cost Reduction Scenario



SCAN

TO UNLOCK PYTHON-POWERED INSIGHTS

This QR code gives you access to an integrated platform built with **Python** that combines interactive **dashboards**, an intelligent **chatbot**, and **scenario-based solutions**.

INSIDE YOU'LL FIND



PYTHON DASHBOARDS

Interactive dashboards



AI CHATBOT

Ask questions, get



SCENARIO SOLUTIONS

Explore what-if scenarios,



SCAN ME





RECOMMENDATIONS

Data-driven actions to reduce energy use, lower emissions, and minimize penalties across NYC buildings.



1. IMPROVE BUILDING PERFORMANCE

Upgrade building systems and adopt energy-efficient technologies to reduce energy consumption.



2. REDUCE EMISSIONS

Implement low-carbon solutions and transition to cleaner energy sources to lower emissions.



3. ENSURE COMPLIANCE

Stay aligned with Local Law 84 & 97 requirements to avoid penalties and maintain compliance.



4. OPTIMIZE COSTS

Identify cost-saving opportunities and optimize energy usage to improve financial performance.



5. LEVERAGE DATA & ANALYTICS

Use data insights and predictive analytics to monitor performance and support smarter decisions.



6. ENGAGE STAKEHOLDERS

Collaborate with building owners, managers, and policymakers to drive sustainable actions together.



These recommendations empower NYC to build a greener, more efficient, and more resilient future.

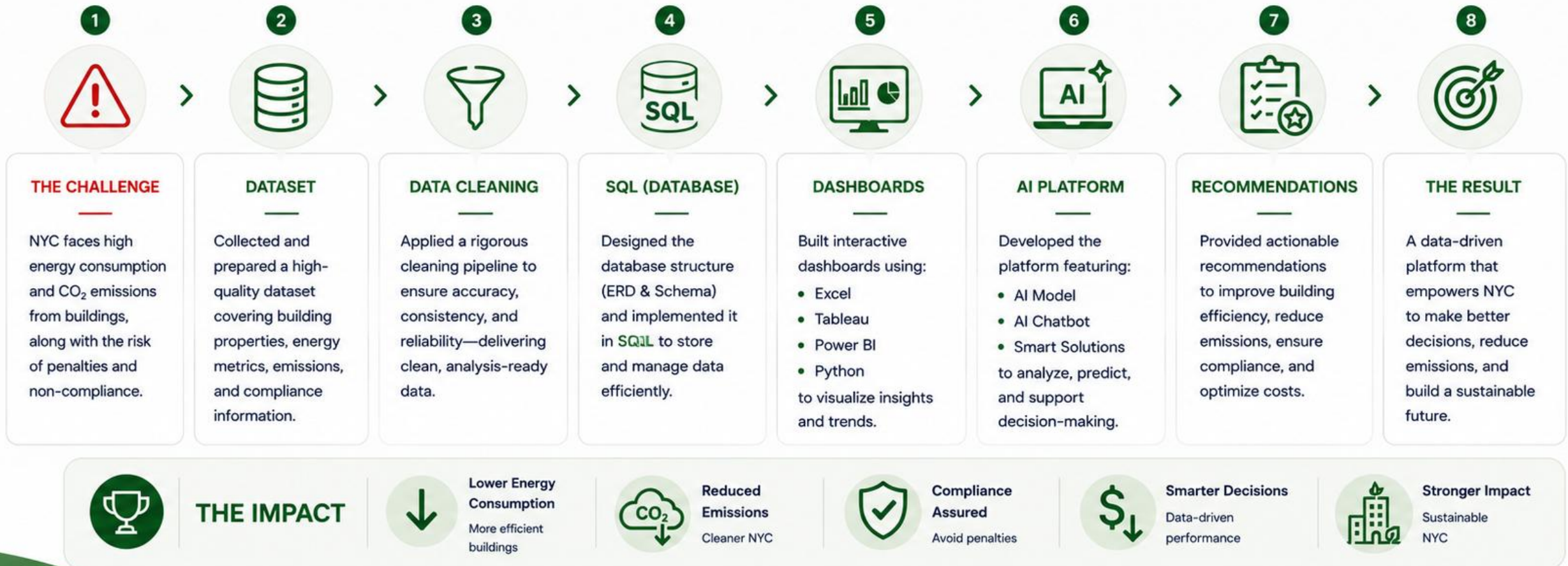




CONCLUSION

From Challenge to Impact

We started with a **challenge**, turned it into a clear **goal**, and followed a **data-driven** journey to turn insights into real **impact**.



From data to decisions, from insights to action, we're building a **greener, more efficient, and more resilient** NYC.

VISIT US ON
GITHUB





THANK YOU

FOR YOUR ATTENTION!

We are happy to answer any **questions**.

